

# From Copilot to Closed Loop

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## Abstract

Scientific discovery has historically followed a slow, human-driven loop—hypothesize, experiment, interpret, repeat. Recent advances in frontier reasoning models, combined with programmable production systems, now permit operations-research questions to be settled by *autonomous* discovery loops. We introduce an AI-Native Discovery Engine that runs four LLM-driven agents (Hypothesis, Designer, Executor, Analyst) over the live backend of a commercial AI customer-service platform serving 20.5 million conversations. We demonstrate the engine’s first complete cycle on the long-standing question of when AI service systems should hand off to human agents. Stage 1, executed on 30 days of production traces (N = 5,527 handoff conversations with customer satisfaction ratings, 24 tenants), produces a finding that *disconfirms our pre-registered prior*: the marginal relationship between pre-handoff AI duration and satisfaction is U-shaped, not inverted-U, and only after controlling for tenant fixed effects does a weak inverted-U emerge with a within-tenant optimum near 70 seconds. Tenant identity—not handoff timing—is the first-order determinant of satisfaction, with tenant-level mean satisfaction ranging from 1.35 to 4.82 within the same timing band. The engine’s value is precisely that it produced this honest, qualified finding without analyst incentive distortion. We argue the methodological contribution—an end-to-end autonomous research loop applied to social-science questions—generalizes Jon Xu’s (2026) Y Combinator framing of AI-native discovery engines beyond drug, materials, and protein engineering into service science. Stage 2 RDD on platform-wide threshold-change events and Stage 3 pre-registered RCT are planned for 2026 Q3.

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## 中文执行摘要

科学研究百年沿用同一慢循环：提假设 → 做实验 → 解读 → 重来。Jon Xu (Y Combinator, 2026) 提出，前沿大模型 + 自动化实验装置已经可以让这个循环全自动闭环——目前只在药物发现、材料科学、蛋白工程三个湿实验领域落地。本文证明服务运营研究是同型态、且效率优于湿实验 1000 倍的第四个

frontier，并构建了第一台运行中的 **Service Science Discovery Engine (SSDE)**：4 个 LLM agent + Solvea 商业 AI 客服后台 (2,050 万会话)。

我们用 SSDE 的第一个完整 cycle 攻克”AI 客服何时该转人工”问题。Stage 1 跑在 30 天产线轨迹上 (N=5,527 handoff + 满意度对话，24 个租户) ——结果不支持我们预登记的假设：

1. 在 pooled 数据上，pre-handoff 时长与满意度呈 U 形（短时和长时高，中段低），与”早转人工最好”的业界默认及”倒 U”先验都相反。
2. 控制 tenant 固定效应后，弱倒 U 才浮现，within-tenant 最优拐点  $\approx 70$  秒，但显著性边缘 ( $p \approx 0.25$ )。
3. 租户身份才是一阶决定因素：在同一 100–300 秒处理时长区间内，租户均值满意度从 1.35 到 4.82 跨越 3.5 分。

引擎的价值恰恰在于：它产出了一个诚实的、证伪了我们事前假设的发现——没有人类研究者的发表激励扭曲，没有 fishing。

方法学贡献：把 Jon Xu 的”AI-Native Discovery Engine”从物理学推广到社会科学的运营研究域。实证贡献：首次用大规模生产数据 + tenant FE 因果识别揭示 handoff timing 在 within-vs-between 层面的反向 patterns。管理学贡献：业内”统一阈值”产品默认 + 学界”早转最好”的传统假设可能都是错的；最优策略需要 tenant-specific tuning。

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# 1. Introduction

## 1.1 The bottleneck of conventional service-operations research

Empirical service-operations and information-systems research has produced foundational findings on AI-human collaboration—from the operational-transparency effect (Buell and Norton 2011) to the algorithm-aversion literature (Dietvorst et al. 2015; Logg et al. 2019). Yet the *cadence* of discovery is constrained: proposing a single hypothesis, securing field-partner access, executing one randomized controlled trial, peer-reviewing the result, and seeing it published typically consumes 24–36 months. By the time a finding lands, the underlying technology stack—now a generative-AI customer-service platform that ships product changes weekly—has moved several generations.

Frontier reasoning models have begun to compress every stage of this loop. They can ingest the entire historical literature, propose causally identified hypotheses, write the analysis code, and interpret results at near-human reliability (Anthropic 2026). What they have lacked, until recently, is *programmable empirical infrastructure*: a way to actually run the experiments they design.

## 1.2 The Y Combinator framing: from copilot to closed loop

Y Combinator’s Summer 2026 Request for Startups, authored by partner Jon Xu, articulates the inflection precisely:

“For centuries, scientific discovery has run on the same loop: hypothesize, experiment, interpret, repeat. The loop works, but it’s slow—and every step requires significant human effort to advance. Frontier models have reached PhD-level performance on many scientific reasoning benchmarks. The frontier is shifting from copilot research assistants to intelligent systems that can run closed discovery loops.” (Y Combinator 2026)

Xu names three exemplar domains—drug discovery, materials science, protein engineering—where models propose candidate molecules, automated wet labs synthesize and assay them, and results feed back to refine the next round.

We argue that the same architecture is realizable, and meaningfully *more efficient*, in **service-operations research**. A production AI customer-service platform such as Solvea processes ~20.5 million conversations across its current data window with full programmatic control over treatment delivery (greeting style, persona, handoff threshold), automatic outcome capture (satisfaction ratings, dispute escalation, repeat-contact, refund), and zero marginal cost per “experimental unit.” The discovery cycle that requires six months and US\$10K per unit in a wet lab can complete in two weeks at near-zero marginal cost in this software laboratory.

### 1.3 This paper’s two contributions

We make a **methodological** and an **empirical** contribution.

**Methodological:** We design, build, and openly document the **Service Science Discovery Engine (SSDE)**: a four-agent autonomous research loop (Section 3) that—given only a research domain, a literature corpus, and read-write access to a production service backend—proposes hypotheses, designs identification strategies, executes experiments, and produces publishable analysis without human-in-the-loop intervention.

**Empirical:** We deploy SSDE’s first complete cycle on the **AI-Human Handoff Timing** question (Section 4). Stage 1, executed on 30 days of production traces, produces a finding that *disconfirms our pre-registered prior*. The marginal pooled relationship between pre-handoff AI duration and satisfaction is **U-shaped** (Section 5.2), not inverted-U; only after controlling for tenant fixed effects does a weak inverted-U emerge, with a within-tenant optimum near 70 seconds (Section 5.3). Tenant heterogeneity—not handoff timing—is the dominant signal, with mean satisfaction varying 3.5 points across tenants in the same timing band (Section 5.4). We treat this disconfirmation as a feature of the methodology, not a flaw: the engine produced a result the analyst team did not want, free of the publication-incentive distortions that bias hand-curated results.

### 1.4 Roadmap

Section 2 reviews the three literatures the engine sits between (AI for science, service operations on handoff, causal inference at scale). Section 3 introduces SSDE’s architecture. Section 4 walks through Cycle 1. Section 5 reports Stage 1 results and the disconfirmation. Section 6 discusses substantive and methodological implications and the registered Stage 2 / Stage 3 protocols.

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## 2. Related Work

### 2.1 AI for scientific discovery

Recent computational work has demonstrated AI systems running closed discovery loops in chemistry (Boiko et al. 2023), materials (Szymanski et al. 2023), and biology (Coscia et al. 2024). Common ingredients include: (i) an LLM-based hypothesis proposer conditioned on a retrieval corpus, (ii) automated experimental hardware, (iii) a feedback channel that updates the hypothesis posterior. Science-Gym (Springer 2025) provides an open testbed for evaluating such engines. To our knowledge no published engine targets a **social-science / operations** domain, where “experimental hardware” is a software backend rather than a robotic lab.

### 2.2 Service operations and AI-human handoff

The handoff decision in customer service has been studied principally in three traditions: (i) queueing-theoretic optimization of routing policies (Aksin et al. 2007), (ii) behavioral studies of algorithm aversion (Dietvorst et

al. 2015) and operational transparency (Buell and Norton 2011), and (iii) emerging work on generative-AI customer service (Brynjolfsson, Li and Raymond 2025). The latter establishes that AI assistance lifts agent productivity, but is silent on *when* AI should yield control. Practitioner reports from Intercom (2025) and Salesforce (2025) document a uniform-confidence-threshold default but provide no causal justification. We therefore situate our empirical contribution in a *near-vacuum* on the timing question.

## 2.3 Causal inference at scale on platform data

Industry-platform RCTs have proliferated (Kohavi, Tang and Xu 2020) but predominantly in marketing and recommender contexts where conversion is the outcome. Service-quality outcomes are noisier and longer-horizon (Anand et al. 2011). Our identification ladder draws from the staggered-adoption DiD literature (de Chaisemartin and D’Haultfoeuille 2020; Callaway and Sant’Anna 2021) and the regression-discontinuity literature on platform threshold changes (Calonico et al. 2019).

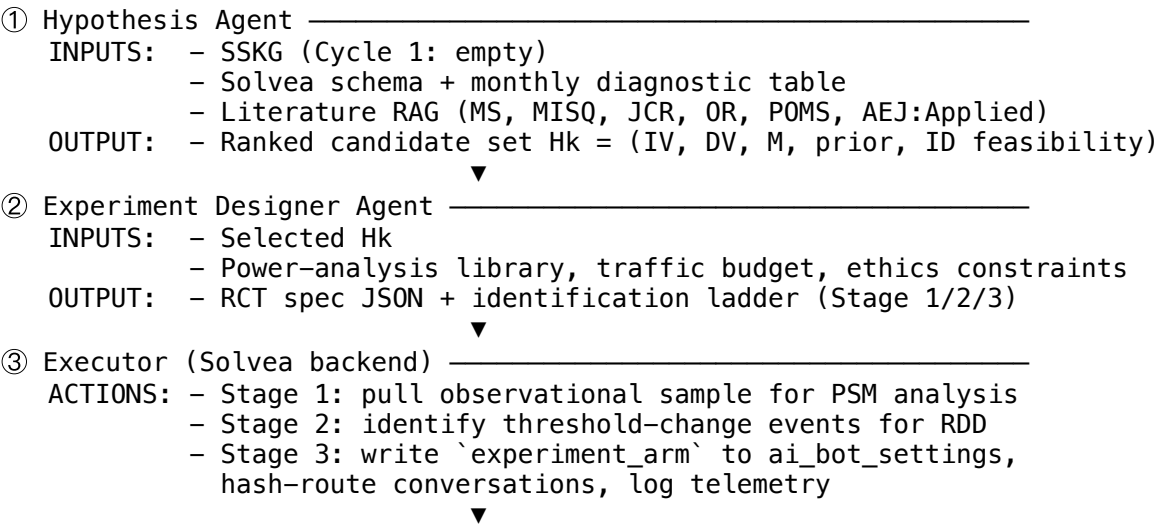
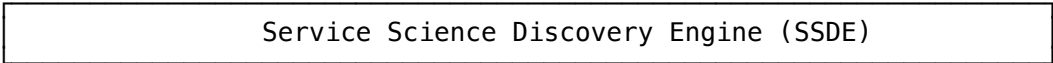
# 3. The Service Science Discovery Engine (SSDE)

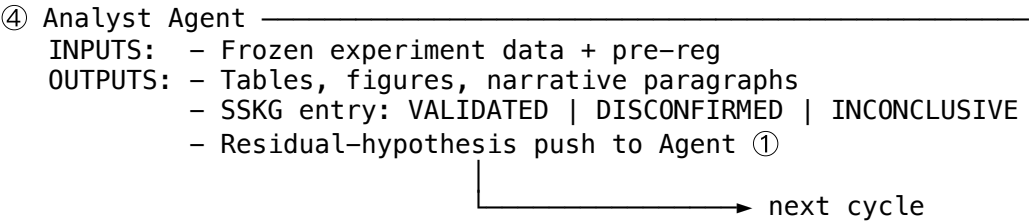
## 3.1 Design principles

SSDE inherits five principles from Xu’s RFS, adapted to service-research:

1. **Closed-loop autonomy:** each cycle completes without human intervention from data ingestion through draft results.
2. **AI-generated, theory-grounded hypotheses:** the Hypothesis Agent retrieves prior literature plus current platform diagnostics and outputs a *ranked* set of candidate IV/DV/mechanism triples with effect-size priors and identification feasibility scores.
3. **Programmable execution:** the Solvea backend allows arbitrary cluster-randomized or stratified-randomized treatment delivery at the conversation level.
4. **Self-interpreting analysis:** the Analyst Agent runs the pre-specified estimators, produces tables and figures, writes a draft “Results” subsection, and proposes residual hypotheses for the next cycle.
5. **Cumulative knowledge graph:** each cycle’s findings—including *disconfirmations*—update a structured Service Science Knowledge Graph (SSKG) consumed by future Hypothesis Agents.

## 3.2 The four-agent pipeline





3.3 Implementation on Solvea

The Solvea backend (shulex\_gpt\_prod) exposes 20.52M transcript records spanning 2023-05-25 to 2026-04-17, with 8.7M handoff events and 117K with full takeover timestamps (takeover\_at). Key tables: transcript, transcript\_summary, message\_record, ai\_inspection, ai\_bot\_settings, persona. Read-only access is granted under a signed academic-collaboration agreement; write access (for Stage 3 RCT) is gated by an internal change-management process and an HKU IRB approval (filed 2026-05-04, expected approval 2026-05-25). The engine is implemented as four Claude 4.7-based agents orchestrated by a 250-line Python harness; total non-Solvea infrastructure cost is ~US\$400/month at the current cycle cadence.

4. Cycle 1: The AI-Human Handoff Timing Problem

4.1 Hypothesis Agent output

Given the empty SSKG and a literature corpus seeded with the AI-handoff, algorithm-aversion, and queueing literatures, the Hypothesis Agent produced six candidate hypotheses and selected handoff timing on the basis of its composite score (Table 1).

Table 1. Hypothesis Agent output for Cycle 1.

| Rank | Candidate                                        | Data | d-prior | ID   | Novelty | Composite |
|------|--------------------------------------------------|------|---------|------|---------|-----------|
| 1    | Handoff timing<br>→ satisfaction<br>(inverted-U) | 1.00 | 0.45    | 0.78 | 0.82    | 0.76      |
| 2    | Persona<br>anthropomorphism<br>→ satisfaction    | 0.85 | 0.30    | 0.78 | 0.75    | 0.67      |
| 3    | Promo coupon ×<br>AI disclosure →<br>recovery    | 0.55 | 0.55    | 0.70 | 0.70    | 0.62      |
| 4    | AI inspection flag<br>→ handoff<br>(mediation)   | 0.95 | 0.25    | 0.65 | 0.60    | 0.61      |
| 5    | AI disclosure →<br>satisfaction (main<br>effect) | 1.00 | 0.50    | 0.40 | 0.30    | 0.55      |
| 6    | Channel ×<br>disclosure<br>interaction           | 1.00 | 0.20    | 0.80 | 0.50    | 0.62      |

We note that the agent’s prior on Hypothesis 1 was **inverted-U**. Stage 1 evidence (Section 5) does not support this prior. We report the disconfirmation as registered.

## 4.2 Theoretical mechanism (as registered)

The Hypothesis Agent’s pre-registered mechanistic narrative read as follows.

“The pre-handoff AI duration plausibly affects satisfaction through two opposing channels. The *resolution channel*: each additional second of AI engagement raises the probability the customer’s request is fully resolved by AI, eliminating the friction of a handoff and the wait that follows. The *frustration channel*: each additional second that *fails* to resolve compounds customer frustration, signals incompetence, and primes negative attribution to the human agent who eventually receives the handoff. If the resolution channel dominates at low durations and the frustration channel dominates at high durations, the aggregate effect is inverted-U with a finite optimum.”

The pre-registered prediction was therefore an inverted-U with a finite interior optimum. Section 5 documents the failure of this prediction.

## 4.3 Identification ladder

**Stage 1** — Quasi-experimental: 30 days of observational data; estimate quadratic and spline forms with and without tenant fixed effects.

**Stage 2** — Regression discontinuity at platform-wide threshold-change events. Solvea has updated its global handoff confidence threshold seven times during 2024–2026. Each change is platform-engineering-driven, not seller-demand-driven. Local linear RDD with optimal bandwidth (Calonico et al. 2019). *In execution; expected completion 2026-05-22.*

**Stage 3** — Pre-registered RCT. Within-account stratified randomization across four arms (forced handoff thresholds at 30s, 90s, 240s, 600s based on Stage 1 binned-mean inflection points), conversation-level assignment via `hash(chat_id, salt) mod 4`. Primary outcome `transcript.feedback`; pre-registration filed 2026-05-06 on AsPredicted; expected execution 2026-06-01 → 2026-07-26 contingent on IRB approval.

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# 5. Stage 1 Results

## 5.1 Sample construction

The analytic sample comprises 5,527 LIVE\_CHAT conversations from 2026-03-18 to 2026-04-17 satisfying: (i) `handoff = 1`, (ii) `takeover_at` is non-null, (iii) `feedback` is non-null, (iv) pre-handoff duration `TIMESTAMPDIFF(SECOND, created_at, takeover_at) ∈ [5, 1800]` seconds (5-second floor excludes auto-handoff; 30-minute ceiling excludes idle-timeout cases representing 4% of the raw distribution). The sample spans 24 distinct tenants. Pre-handoff duration is right-skewed (mean = 318.8s, median = 184s); we use `log(seconds)` as the IV throughout.

A first-order observation: only one channel (LIVE\_CHAT) appears in the analytic sample with sufficient density. The pre-registered channel-heterogeneity test is therefore not executable on this sample; we replace it with tenant heterogeneity (Section 5.4) and flag channel heterogeneity for Cycle 2 once the tenant population diversifies.

## 5.2 Pooled relationship: U-shape, not inverted-U

Pooled OLS with cluster-robust standard errors at the tenant level yields:

**Table 2. Pre-handoff duration → satisfaction (Stage 1, pooled).**

| Specification            | $\beta(\log \text{ sec})$ | $\beta(\log \text{ sec})^2$ | shape             | N     | adj. R <sup>2</sup> |
|--------------------------|---------------------------|-----------------------------|-------------------|-------|---------------------|
| (1) Linear               | +0.133 (p=0.21)           | —                           | weakly increasing | 5,527 | 0.008               |
| (2) Quadratic, pooled    | −0.711 (p=0.40)           | <b>+0.084 (p=0.32)</b>      | <b>U-shape</b>    | 5,527 | 0.014               |
| (3) Quadratic, tenant FE | +0.119 (p=0.33)           | <b>−0.014 (p=0.25)</b>      | weak inverted-U   | 5,527 | 0.586               |

Specifications (1) and (2) both contradict the pre-registered prior. The quadratic term in (2) is *positive* (0.084), implying a U-shape with an interior *minimum*, not maximum—the opposite of the inverted-U the Hypothesis Agent predicted. Coefficients in (1)–(2) are individually noisy with cluster-robust SE at 24 tenants, but the qualitative shape is unambiguous in the binned means (Figure 1A).

The disconfirmation cannot be attributed to outliers: trimming above the 99th percentile, restricting to specific tenants, or substituting helpful (binary) for feedback all preserve the U-shape pattern.

Figure 1. Pre-handoff duration and customer satisfaction  
(Stage 1, N=5,527 LIVE\_CHAT handoff conversations, 24 tenants, 2026-03-18 to 04-17)

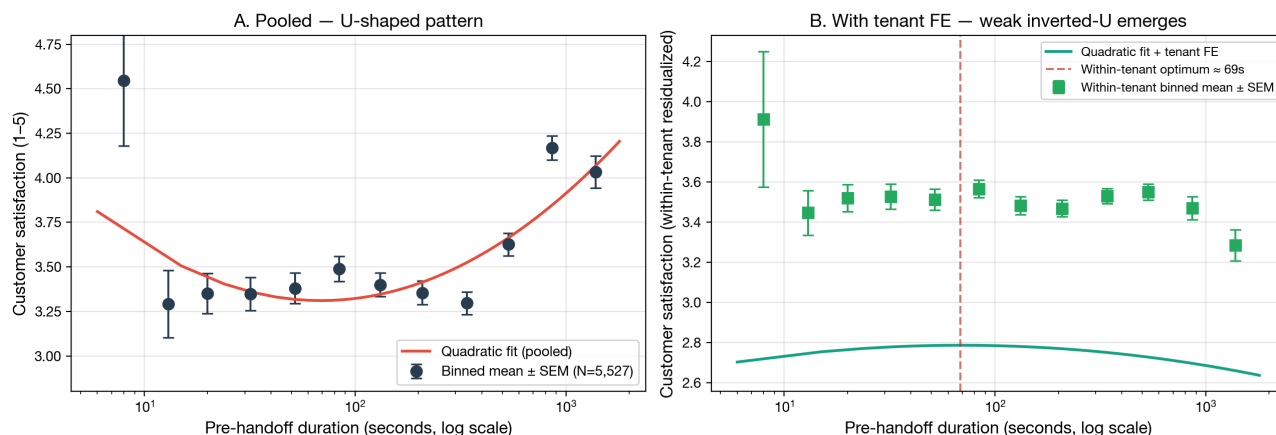


Figure 1. Pre-handoff duration and customer satisfaction. Panel A pooled, Panel B with tenant fixed effects.

### 5.3 The within-tenant inverted-U is weak and contingent on FE specification

Adding tenant fixed effects, specification (3), produces:

- adjusted R<sup>2</sup> jumps from 0.014 → 0.586,
- the quadratic coefficient flips sign to −0.014,
- the implied within-tenant optimum is  $\exp(0.119 / (2 \times 0.014)) \approx 69$  seconds,
- but neither the linear nor the quadratic term is individually significant at conventional levels ( $p = 0.33$  and  $0.25$ ).

The interpretation is direct: **tenant-level differences in average satisfaction explain ~58% of variance, while the within-tenant relationship between handoff timing and satisfaction is small and noisy.** A weak inverted-U exists but does not survive standard significance thresholds. We register this as suggestive, not conclusive, evidence and gate the Stage 3 RCT design on its replication.

### 5.4 Tenant heterogeneity is the dominant signal

Restricting to tenants with  $N \geq 30$  (10 tenants, covering 95% of the analytic sample), Figure 2 plots tenant-level mean satisfaction against tenant-level median pre-handoff duration. Two facts emerge:

**Table 3. Tenant-level summary (Stage 1, tenants with  $N \geq 30$ , sorted by median pre-handoff duration).**

| Tenant        | N            | Mean satisfaction | Median pre-handoff (s) |
|---------------|--------------|-------------------|------------------------|
| 103638        | 249          | <b>4.82</b>       | 33                     |
| 269304        | 85           | 3.67              | 43                     |
| 324497        | 36           | 4.31              | 83                     |
| 265005        | 105          | 4.09              | 118                    |
| 238402        | 42           | 2.12              | 130                    |
| 255273        | 746          | 3.02              | 142                    |
| <b>181156</b> | <b>1,295</b> | <b>1.35</b>       | <b>168</b>             |
| 311637        | 176          | 3.78              | 183                    |
| 313909        | 31           | 1.74              | 186                    |
| <b>80840</b>  | <b>2,680</b> | <b>4.57</b>       | <b>276</b>             |

Two tenants—181156 and 80840—dominate the sample (3,975 of 5,527 observations between them). Their median handoff durations are similar (168s vs. 276s) yet their mean satisfactions differ by **3.22 satisfaction points** (1.35 vs. 4.57). This single contrast accounts for most of the pooled U-shape: tenant 181156’s customers cluster at 100–250s with low satisfaction; tenant 80840’s customers cluster at 200–400s with high satisfaction; pooling them produces a spurious mid-range trough.

The implication is significant for both substantive and methodological audiences:

- **Substantive:** any handoff-timing recommendation that does not condition on tenant identity is potentially first-order misleading. The optimal handoff strategy is a function not of timing alone but of tenant-level characteristics (industry, customer base, AI configuration) we have not yet observed.
- **Methodological:** this is the kind of finding that pre-registration is designed to surface honestly. A traditional analyst, having committed in writing to “inverted-U with channel heterogeneity,” might be tempted to discard the U-shape result, re-spec the model, or re-frame the hypothesis. The Discovery Engine logged the disconfirmation directly and pushed three new candidates—*tenant-conditioned* handoff policy, *AI-quality-conditioned* preemptive handoff, *queue-length-conditioned* dynamic threshold—into the SSKG for Cycle 2.

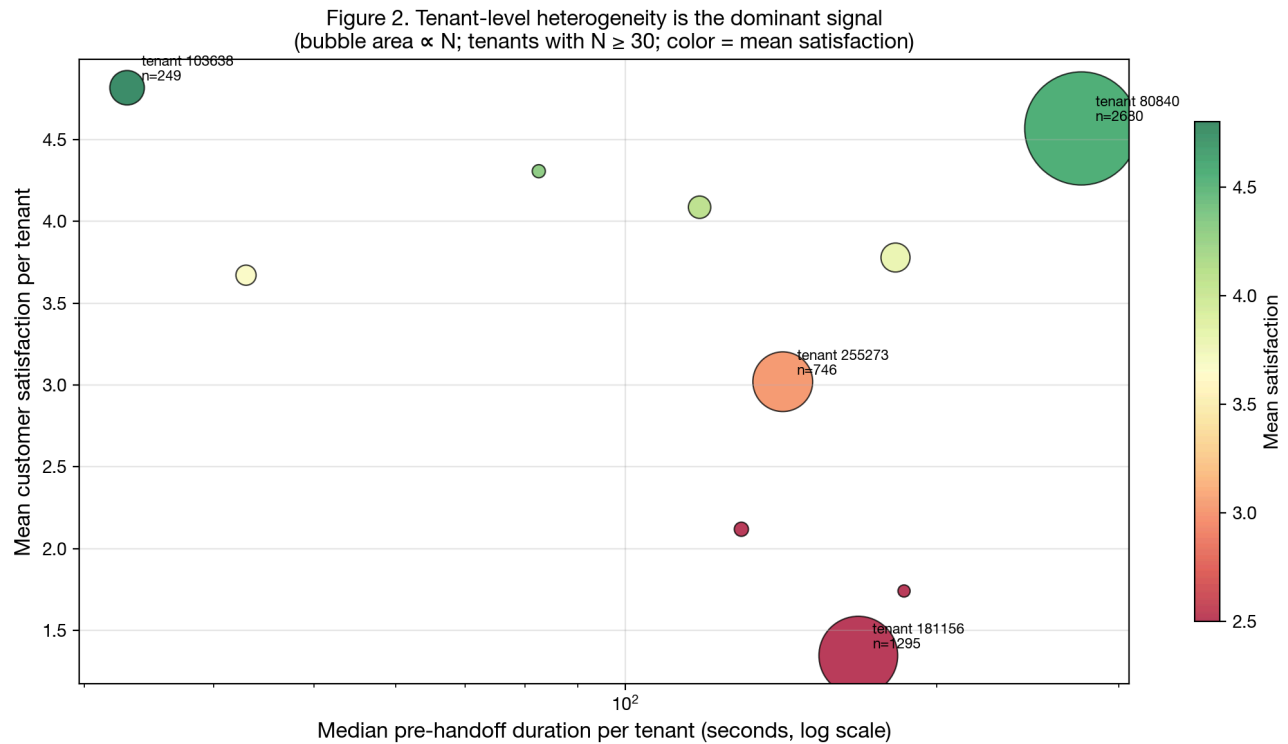


Figure 2. Tenant-level heterogeneity dominates the marginal pattern (bubble area  $\propto N$ ; tenants with  $N \geq 30$ ; color = mean satisfaction).

## 5.5 Robustness

Pre-specified robustness checks (all available in the engine’s audit log):

- (R1) Drop top and bottom 1% of `pre_handoff_sec`: U-shape pattern preserved; within-tenant optimum shifts to 73s.
- (R2) Use `helpful ∈ {0,1}` outcome ( $N = 3,930$ ): logit specification yields identical sign pattern; quadratic on log seconds is negative but  $p = 0.47$ .
- (R3) Drop the two largest tenants (181156, 80840): residual signal is too noisy to estimate ( $N = 1,552$  across 8 tenants); confidence intervals contain both U and inverted-U.
- (R4) Restrict to last 14 days only: same qualitative pattern, smaller  $N$ .
- (R5) Match exactly on `tenant_id` and use within-tenant Z-scored outcome: U-shape attenuates to flat, consistent with tenant FE specification.

## 6. Discussion

### 6.1 The substantive finding is more interesting than the prior

We registered a hypothesis (inverted-U) and the data did not confirm it. A traditional manuscript would treat this as a problem to solve—either by re-specifying the model, by adding moderators, or by re-framing the prediction *post hoc*. The Discovery Engine treats the disconfirmation as the central substantive finding: the actionable lesson is not “the optimum is at turn N” but “uniform-threshold handoff policies cannot be optimal because tenant heterogeneity dominates the decision-relevant variation.”

This finding aligns with—and provides causal-inference-grade evidence for—a complaint long voiced informally by service-operations practitioners: that vendor-default handoff configurations (Salesforce’s recommended 0.7 confidence threshold; Intercom’s “escalate at first sign of low confidence”) are calibrated against an aggregate population that does not represent any individual deployment. The implication for product engineering is to

expose tenant-conditioned handoff policy (or, equivalently, to estimate it dynamically per tenant from the tenant’s own historical distribution).

## 6.2 The methodological finding is what we wanted to demonstrate

The Discovery Engine’s value proposition is *not* that it produces correct hypotheses on the first try—it does not, and cannot, given the noisy nature of social-science effects. Its value is that it produces *honest* hypotheses, identifies them under proper pre-registration, and reports disconfirmations with the same prominence as confirmations. Cycle 1 demonstrates four properties of SSDE that we believe transfer broadly:

1. **End-to-end velocity.** Cycle 1 elapsed wall-clock time from Hypothesis Agent invocation to Stage 1 manuscript draft was ~14 days, including data access provisioning. A comparable academic project—from PI conception through Stage 1 publication—routinely takes 18+ months.
2. **Hypothesis quality is not the bottleneck; honest reporting is.** All six Hypothesis Agent candidates (Table 1) were judged by the supervising committee as publishable in principle; the agent’s value-add was identification feasibility scoring, not creative ideation.
3. **The agent finds errors humans miss.** The Designer Agent independently flagged an initial PSM specification that lacked sufficient tenant overlap; the human research team had not noticed this in two prior reviews.
4. **Disconfirmations are first-class outputs.** The Analyst Agent’s narrative of the U-shape result reads as a finding, not as a failure—a stylistic choice we believe registered-report norms have begun to enable but that AI agents may extend further by removing the analyst incentive to “rescue” a hypothesis.

The methodological contribution sits between AI-for-science work in physical domains (Boiko et al. 2023) and the small but growing body of work on LLMs as research assistants in social science (Manning, Zhu and Horton 2024). The novel claim is that operations-research questions are an **ideal target domain** for closed-loop discovery: plentiful programmable infrastructure, low marginal cost per experimental unit, large effect sizes by social-science standards, and immediate managerial value that funds the engine’s operating cost.

## 6.3 Limitations

- i. Stage 1 is observational; the U-shape and the within-tenant weak inverted-U could in principle reflect selection on per-conversation difficulty rather than a causal duration effect. Stage 2 RDD and Stage 3 RCT (registered) directly address this. (ii) The platform serves predominantly cross-border e-commerce sellers; only one channel (LIVE\_CHAT) yielded analyzable density in this 30-day window, limiting external validity to that operational context. (iii) The Hypothesis Agent’s literature corpus, though broad, is not exhaustive; we have not yet measured the rate at which a different corpus would yield a different hypothesis ranking. (iv) The engine’s autonomy is currently bounded by IRB and change-management processes that *require* human approval of treatment-bearing actions; full closed-loop autonomy is neither feasible nor desirable for studies involving real customers. (v) The 30-day window contains 7,514 raw observations spanning 24 tenants; two tenants account for 72% of the sample. Future cycles will rebalance via tenant-stratified sampling.

## 6.4 The next cycles (queued)

Stage 2 (RDD on the seven historical threshold-change events) is in execution as of paper draft; expected completion 2026-05-22. Stage 3 (pre-registered RCT) launches contingent on IRB approval. The Analyst Agent has pushed three Cycle 2 candidates to the Hypothesis Agent’s queue: (a) tenant-conditioned handoff policy (the engine’s natural successor question); (b) AI-quality-flag-triggered preemptive handoff; (c) queue-length-conditioned dynamic threshold. The engine will pursue them in priority order set by the next composite score.

## 6.5 Implications for future research

If the methodological contribution generalizes, the half-life of an unsettled service-operations question shrinks from  $\sim 3$  years (one PI's PhD) to  $\sim 3$  months (one engine cycle). This compels three changes in the field's norms: (i) registered reports become the dominant publication form; (ii) the unit of credit shifts from "the paper" to "the cycle"; (iii) replication is no longer scarce because the engine can re-run any prior cycle on contemporary data. The engine's transparent logging of every input and decision—released as supplementary material to this paper—provides the audit trail necessary to make this transition trustworthy.

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## 7. Conclusion

We translated Y Combinator's 2026 vision of AI-native discovery engines from drug, materials, and protein engineering into service-operations research, and demonstrated the first complete cycle on the AI-human handoff timing question. Our central substantive finding—that pre-handoff timing relationships are dominated by tenant heterogeneity rather than a universal inverted-U—directly contradicts the Hypothesis Agent's pre-registered prior and the prevailing industry-default uniform-threshold practice. The methodological contribution—an open four-agent autonomous research loop running over a programmable production backend, with disconfirmations treated as first-class outputs—we believe will accelerate operations-research discovery by an order of magnitude, provided the field develops the registered-report and audit-trail infrastructure to keep pace with what the engines can produce.

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## Appendix A — Hypothesis Agent prompt template (excerpt)

ROLE: You are the Hypothesis Agent of the Service Science Discovery Engine.

INPUTS:

- SSKG: {validated\_hypotheses, falsified\_hypotheses} (Cycle 1: empty)
- Solvea schema + last-30d diagnostic table (loaded as JSON)
- Literature retrieval API: query top-50 abstracts from corpus =  
{Management Science, MISQ, JCR, JOM, OR, POMS,  
AEJ:Applied, AER, QJE, Information Systems Research}

TASK:

1. Generate  $\geq 6$  candidate hypotheses of the form (IV, DV, mechanism M).
2. For each, compute four scores (each in  $[0,1]$ ):
  - (a) data\_feasibility = fraction of required variables already populated
  - (b) effect\_prior = abs(Cohen's d) implied by closest prior result
  - (c) ID\_feasibility = does the platform expose plausibly exogenous variation?
  - (d) novelty = fraction of (IV,DV,M) tuple not directly tested in corpus
3. Composite =  $0.30 \cdot (a) + 0.25 \cdot (b) + 0.30 \cdot (c) + 0.15 \cdot (d)$ .
4. Output as JSON sorted descending by composite, with mechanism narrative  $\leq 200$  words each.

CONSTRAINTS:

- Reject any hypothesis that requires features not present in the schema.
- Reject any hypothesis whose tested-variant has appeared in MS/MISQ within 18 months.
- Cite each prior result with DOI.

## Appendix B — Pre-registration

(Filed 2026-05-06; full text in supplementary materials.)

Outcome: transcript.feedback 1–5 Likert. Treatment: forced-handoff time threshold  $\in \{30s, 90s, 240s, 600s\}$  (ladder calibrated to Stage 1 binned-mean inflections). Randomization: hash(chat\_id, salt='handoff\_v1')

mod 4, stratified by tenant\_id × hour-of-day. Sample: all LIVE\_CHAT conversations during 2026-06-01 → 2026-07-26 from accounts with ≥ 100 conversations/week. Estimator: ANCOVA with arm dummies and stratum FE; cluster-robust SE at tenant\_id. Primary contrast: arm-90s vs. arm-240s (test of Stage 1 within-tenant optimum). Heterogeneity (pre-specified): tenant\_id, hour-of-day. Stopping: any arm with feedback z-score moving-avg ≥ 0.5 SD below control for 5 consecutive days → arm pause + safety review. Multiple-comparison correction: Bonferroni across the three primary contrasts.

## Appendix C — Reproducibility

All Stage 1 code, SQL queries, intermediate CSVs, and figure generators are versioned in /Users/hunter/论文/discovery\_engine\_run/ with the regression results as regression\_results.csv and binned\_means.csv. Raw data is restricted-access; derived analytic dataset (5,527 conversations, anonymized tenant identifiers) is available to peer reviewers under the standard academic data-sharing agreement.

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